

A 3D-Printed Touch Sensor with a Single-Stroke Conductive Path

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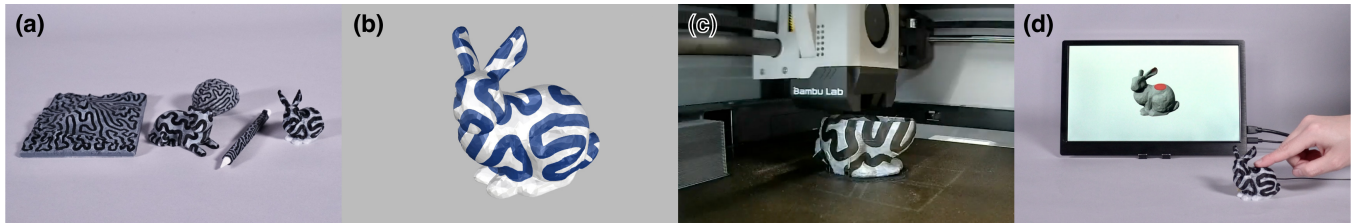


Figure 1: Overview of the proposed sensor: (a) users can embed touch sensing capabilities into any existing 3D model. (b) A single-stroke conductive path is designed and modeled. (c) The sensor is fabricated as a single unit using two types of materials, conductive and non-conductive, through 3D printing. (d) Touch interactions at any point can be detected with a single wire, and the input points can be redefined even after fabrication.

Abstract

We present a novel 3D-printed touch sensor integrating a single-stroke conductive path, enabling touch detection at any point on three-dimensional surfaces with just a single wire. By utilizing Swept Frequency Capacitive Sensing (SFCS), this sensor provides accurate detection while allowing redefinition of input points even after fabrication. Fabricated using conductive and non-conductive materials in a single process, the sensor overcomes limitations of existing methods that require extensive wiring or predefined touch areas. We conducted evaluations to demonstrate the sensor's high accuracy in distinguishing touch points. Additionally, we explored its potential through applications such as interactive toys, educational tools, and customizable interfaces.

Keywords

3D printing; touch sensor; fabrication; user interfaces

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1 Introduction

3D printing technology can produce versatile three-dimensional shapes as a single unit and is being widely applied, from creating objects that fit existing ones to fabricating complex mechanisms [20]. In recent years, methods for incorporating sensing capabilities into

3D-printed objects by utilizing materials and structures have also been proposed. In particular, as Willis et al. [25] and Tejada et al. [21] have suggested, attention is focused on technologies that integrate functionality with 3D printing, enabling the creation of sensors ready for use immediately after printing, without the need for assembly or complex processes.

Among sensors, touch sensing, which detects contact with objects, has been widely studied and proposed as a fundamental aspect of interaction [2, 8, 24]. However, existing methods for detecting touch points on three-dimensional shapes often require pre-designing specific touch areas [9, 18]. This approach limits the number of detectable points and prevents post-fabrication modifications, reducing flexibility in usage. To enable highly flexible touch detection across multiple areas, extensive wiring is usually required [6, 11, 13]. This not only prevents single-piece fabrication and complicates the manufacturing process but also may interfere with interaction. Although there is research where swept-frequency capacitive sensing (SFCS) has been utilized to enable touch point detection with a single wire [23], challenges remain in distinguishing points that are equidistant from the wiring location.

In this study, we propose a novel 3D-printed sensor capable of detecting touch at arbitrary points by creating a conductive path that runs in a single stroke across a three-dimensional surface and applying SFCS at its endpoint, enabling touch detection at any point with a single wire.

The contributions of our research are summarized below:

- Proposal of a system that enables touch detection at any point on any three-dimensional shape using a single wire with a single-stroke conductive path.
- Application of SFCS to 3D-printed sensors with conductive materials, creating a sensor where touch input points can be redefined and adjusted as needed, even after fabrication.
- Evaluation of the sensing accuracy of the single-stroke conductive path.

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- Development of application examples utilizing the proposed sensor to demonstrate its potential for various uses.

2 Related Work

While methods such as attaching copper tape [17, 22] or injecting conductive paste after fabrication [16] have been proposed to add sensing capabilities to 3D-printed objects, recent advancements in multi-material FDM (Fused Deposition Modeling) 3D printing with conductive filaments have made it possible to create interactive objects with embedded sensing functionalities in a single fabrication process [5, 14, 19].

For touch sensing technologies on three-dimensional curved surfaces, methods such as using pneumatic sensing by creating holes at the detection points [21] or using conductive materials at the detection areas [1, 9, 18] have been proposed. However, these approaches have the limitation that the interactive areas are pre-determined during the design phase and cannot be modified after the sensor is created. Additionally, in methods involving the design of conductive areas, increasing the number of detection points often requires more wiring or the use of large external sensing components.

To enhance the flexibility of touch position detection, methods have been proposed that apply mutual-capacitive sensing, a technique also used in touchscreens, to three-dimensional curved surfaces [6, 11, 13]. However, these methods involve extensive wiring, making fabrication complex and rendering the integration of interactive functionalities impractical. Similarly, sensors utilizing Electrical Impedance Tomography (EIT) face the challenge of requiring a large amount of wiring [26].

In this study, we focus on SFCS as a method to enable highly flexible touch detection with a single wire. SFCS is a technology that detects touch interactions by leveraging the fact that the current path through a material varies with frequency. It sweeps multiple frequencies of alternating current voltage and detects touch gestures based on impedance feedback. In *Touché* [15], touch interactions with various objects were enabled by classifying impedance changes using Support Vector Machine (SVM), a type of machine learning. SFCS has been applied to a wide range of materials, including plants [12], conductive fabrics [4], and soft conductive materials [23]. Inspired by these prior studies, this research applies SFCS to 3D-printed conductive objects. To address the challenge of distinguishing touch points equidistant from the wiring, we designed a single-stroke conductive path, solving this problem effectively

3 Single Stroke Sensor

3.1 Surface-Filling Single Stroke Curve

To evenly distribute a single-stroke conductive path across a three-dimensional curved surface, we utilized a surface-filling curve. A surface-filling curve is a continuous mapping from a one-dimensional space onto a two-dimensional surface, ensuring that every point on the surface lies within a specified maximum distance from the curve. We considered this type of curve to be suitable for designing conductive paths on 3D objects.

In this work, we utilized the software proposed by Noma et al. [10] to generate surface-filling curves. The method's robustness and interactive features, such as allowing users to explore and

modify curves or adjust parameters like density spatially led us to consider it well-suited for applications in digital fabrication. In this system, the initial curve can be specified using vertex indices. In this study, instead of using a closed curve, an open curve was used as the input curve to generate a surface-filling curve, resulting in a single-stroke surface-filling curve with both a starting and an ending point.

3.2 Sensing Principle

This sensor adopts SFCS, similar to *foamin* [23], to detect touch points based on impedance changes across multiple frequencies. Impedance is a combination of the sensor's resistance and the touch surface's capacitance. The resistance is determined by the distance from the wiring to the touch point, while the capacitance depends on the contact area of the finger. In this study, by designing a single-stroke conductive path extending from the wiring point, the distance from the wiring to any touch point is ensured to be unique, enabling highly accurate detection of touch points (Fig. 2).

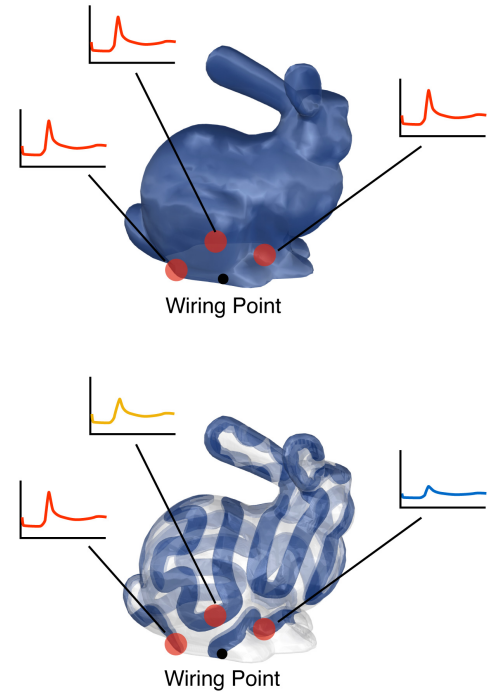


Figure 2: In SFCS, touch points located at similar distances from the wiring tend to produce similar waveforms, making it difficult to distinguish between them (top). By using a single-stroke conductive path, the distance from the wiring to each touch point becomes unique, resulting in distinct waveform data and enabling accurate distinction (bottom).

4 Fabrication

The proposed Single Stroke Sensor can be fabricated as a single unit using a commercially available FDM 3D printer and conductive

filament. This chapter describes the fabrication process, including 3D modeling, 3D printing, and thermal press fitting, as well as the configuration of the sensor board.

3D Modeling. Using the surface-filling curve generated with the aforementioned software, Boolean operations were performed to create 3D data for printing. For remeshing and simplification of the mesh data used in the surface-filling curve software, mesh processing and modeling tools such as MeshLab and Metasequoia were utilized. Rhinoceros, along with its plugin Grasshopper and the volumetric modeling plugin for Grasshopper, Dendro, was used for size adjustments and Boolean operations.

3D Printing. The proposed sensor can be fabricated using any 3D printer capable of printing with two types of materials: conductive and non-conductive. For this study, we used the dual-nozzle TENLOG TL-D3 and the Bambu Lab X1 Carbon equipped with an Automatic Material System (AMS). As conductive materials, we employed NinjaTek's Eel TPU filament and ProtoPasta's Conductive PLA filament.

Thermal Press Fitting. Contact pins were inserted into holes of connector parts through thermal press-fitting using a soldering iron. This process securely bonded metal to conductive printed material, ensuring reliable electrical conduction between them.

Sensing System. For the implementation of the SFCS system, there are two approaches: one utilizing analog waves generated by a programmable single-scan waveform generator [15], and the other employing PWM through an LC circuit [7]. In this study, we adopted the latter approach, which can be implemented on a general-purpose microcontroller board. An Arduino Uno with a clock frequency of 16 MHz was used, and the implementation was based on previous studies [7]. The frequency sweep was performed between 2 kHz and 5 kHz. For waveform classification, Support Vector Machine (SVM) was employed, using the SVC function from the Python library scikit-learn. The overall system overview is shown in Fig. 3.

5 Evaluation

We conducted an experiment to evaluate the detection accuracy of touch points with the proposed sensor. To examine the influence of the single-stroke conductive path, we 3D-printed two experimental samples: one made entirely of conductive material and the other employing the proposed single-stroke conductive path. Both samples were based on the Stanford Bunny model, had identical dimensions and thickness, and were fabricated using TPU.

The experiment was conducted on a wooden desk, with a chair used for seating and both feet placed flat on the floor. The four detection points used for evaluation are shown in Fig. 4. Training data were collected using a data acquisition program operating at 5 fps. For each detection point, data collection was repeated three times, with each session lasting 4 seconds, resulting in a total of 60 training data samples per point. After data collection, classification accuracy was evaluated using an SVM. For each detection point, touch input was performed 10 times, with 10 classification outputs generated per touch, yielding a total of 100 classification results per point. Since the input was considered unstable immediately

after the start and just before the end of each touch, 10 stable classification outputs from the middle of the input were used as the results. The detection accuracy results for each sample are shown in Table 1.

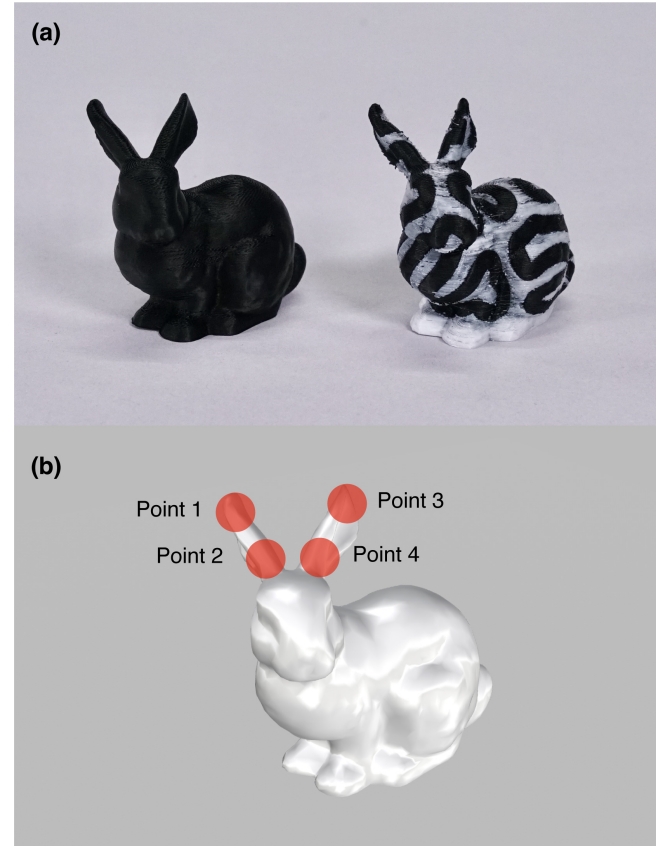


Figure 4: (a) The Stanford Bunny models 3D-printed as evaluation samples: a wholly conductive model and a single-stroke conductive model. (b) The four touch points used as input for the evaluation.

The experimental results revealed that for the wholly conductive sample, the classification was unstable at all evaluation touch points, resulting in low accuracy. By comparison, the sample with the single-stroke conductive path achieved stably accurate classification results at all evaluation touch points. These findings confirmed that by placing a single-stroke conductive path on a three-dimensional curved surface, it becomes possible to accurately distinguish areas with similar distances from the wiring, which were difficult to differentiate when the entire surface was uniformly conductive.

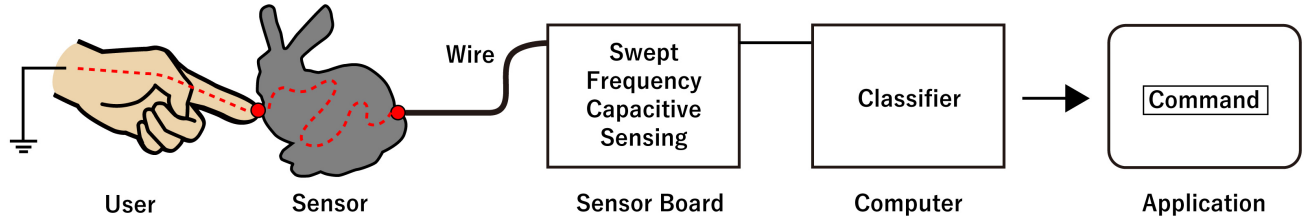


Figure 3: An overview of the system architecture. Waveform data obtained from SFCS touch inputs are classified using a classifier, and the resulting data are utilized in applications.

Table 1: Accuracy of touch input detection for each experimental sample.

		Wholly Conductive				
		Predicted				
Actual		No Touch	Point 1	Point 2	Point 3	Point 4
	No Touch	100	0	0	0	0
	Point 1	0	28	2	70	0
	Point 2	0	41	8	51	0
	Point 3	0	18	0	51	31
	Point 4	0	12	0	38	50

		Single-Stroke Conductive				
		Predicted				
Actual		No Touch	Point 1	Point 2	Point 3	Point 4
	No Touch	100	0	0	0	0
	Point 1	0	100	0	0	0
	Point 2	0	0	100	0	0
	Point 3	0	0	0	100	0
	Point 4	0	0	0	0	100

6 Application

This sensor can be fabricated in arbitrary three-dimensional shapes, making it adaptable to a variety of objects. With only one wire required, it does not interfere with interactions, and its input points can be redefined. Leveraging these features, we created four application examples (Fig. 1 (a)).

6.1 Interactive Cat Toy

This sensor can be fabricated in various shapes, allowing the creation of toys in the form of animals or characters. Since cats have areas they enjoy being touched and areas they dislike, we created an interactive toy that changes its response based on the area being touched (Fig. 5 (a)).

6.2 Educational Brain Model

The use of 3D-printed models in anatomical education has gained attention as an alternative to traditional cadaveric dissection, offering advantages such as customizability, affordability, and new learning possibilities [3]. To explore this potential, we applied the proposed sensor to a brain model (Fig. 5 (b)). Since the input points can be redefined, we implemented two examples: one for learning the names of brain regions and another for learning their functions.

6.3 Mode-Switching Pen Grip

We integrated the proposed sensor into a 3D-printed grip model designed to fit an existing pen, enabling an application that switches drawing modes based on how the pen is held (Fig. 5 (c)). As an example of user-customizable functionality, we developed two scenarios: one where the drawing color changes depending on the surface being touched, and another where the drawing pen style changes based on the height of the grip position.

6.4 Tangible Topological Map

The Tangible Topological Map is a tactile map that allows users to experience features such as terrain elevations through touch (Fig. 5 (d)). We applied the proposed sensor to a 3D model that recreates the topography of Mount Fuji and its surrounding areas. An application was implemented where touching different parts of the model displays corresponding descriptions on a map shown on the screen.

7 Conclusion and Future Work

We proposed a 3D-printed touch sensor with a single-stroke conductive path, enabling accurate and reconfigurable touch detection on three-dimensional surfaces using a single wire. Our evaluations demonstrated its reliability, and its potential was highlighted through various applications, paving the way for more versatile and interactive interfaces.

Future work includes investigating the relationship between conductive path parameters, such as line width and spacing, and

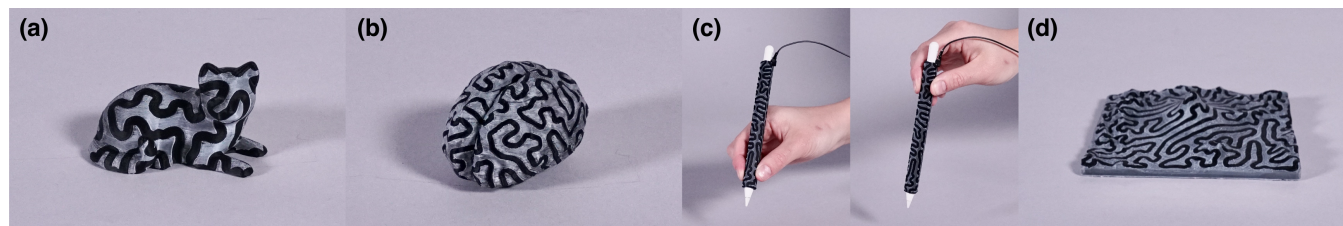


Figure 5: Application examples: (a) Interactive Cat Toy (b) Educational Brain Model (c) Examples of how to hold the Mode-Switching Pen Grip (d) Tangible Topological Map

sensing performance. Additionally, developing a method to predict and detect intermediate touch points by training on a few input points, as well as enabling multi-touch detection by applying SFCS to both endpoints of the conductive path, are key directions to explore.

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